Nexus: Egocentric Annotation Guidelines

Project Overview

In this project, you will label videos to describe what a person (the Operator) is doing with their hands while they complete a task.

[Example Annotation Loom]

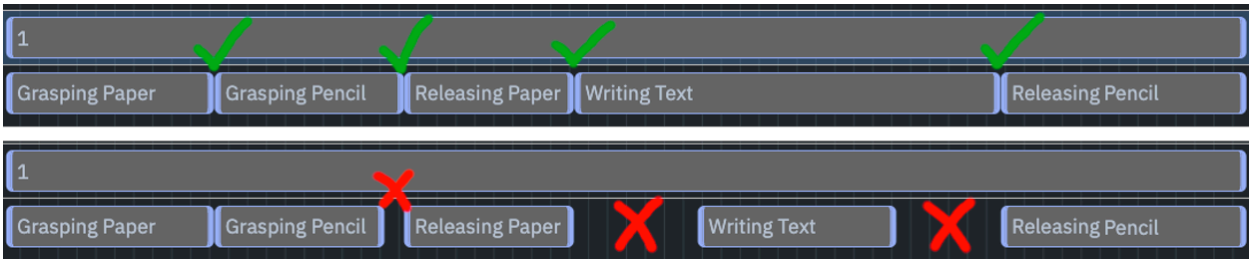
● Action

An ● **Action** label describes what the operator's hands are doing. ● **Action** labels are atomic, occur back to back with no space between them, and take the form of "<verb>ing <noun>"

General Labeling Do's and Don't's

- **DO** use "Verb + Object" phrasing on labels
  - ✅ Grasping Pen, Writing Text, Releasing Pen
  - ❌ Getting Pen from Drawer, Writing, Pen is Put Away
- **DO** use consistent object names and verbs in your labels
  - ✅ Grasping Knife, Grasping Apple, Releasing Knife, Releasing Apple
  - ❌ Grasping Knife, Grabbing Apple, Releasing Blade, Dropping Fruit
- **DO** use concise and specific language
  - ✅ Apple, Fork, Shirt,
  - ❌ Fruit, Utensil, Clothing
- **DO** Accurately Mark the start and end frame for each label
  - ✅ Label start/end frame < 0.5 seconds offset from the actual event.
  - ❌Label start/end frame > 0.5 seconds offset from the actual event.
- **DON'T** Include ● **Action**s that aren't observable or related to the hands
  - ✅ Erasing Whiteboard, Installing RAM, Cutting Paper
  - ❌ Considering options, Adjusting Camera, Listening to Music

- **DON'T** include gaps between ● **Action** labels



- **DON'T** qualify nouns with generic articles in ● **Action** labels
  - ✅ Grasping Towel, Drying Hands, Releasing Towel
  - ❌ Grasping **a** Towel, Drying **her** Hands, Releasing **the** Towel

When labeling, if an object cannot be clearly identified throughout the video, describe its size, shape and/or color. Prefer descriptive nouns over generic ones and stay consistent by always referring to the same object with the same descriptors. For example:

- Actual: Grasping Adhesive, Slicing Grape
- ✅ Grasping Short Blue Bottle, Slicing Round Purple Fruit
- ❌ Grasping Small Cylinder, Slicing Purple Sphere

Failure & Recovery

What Is a Failure?

A **failure** occurs when the gripper attempts an action but does not successfully complete it as intended.

Examples include:

- Missing an object during a grasp
- Dropping an object immediately after lifting
- Slipping while holding an item
- Placing or inserting an object incorrectly
- Losing control of an object during movement

If the intended outcome is not achieved, the action must be marked as a failure.

What Is a Recovery?

A **recovery** occurs when the gripper corrects a previous failure by attempting the intended action again.

Examples:

- Re-grasping an item after dropping it
- Reattempting placement
- Adjusting a misplaced object
- Repeating an action after a miss

Each recovery attempt must be labeled as a **new, separate action**.

Important Recovery Rules

- A recovery must always follow a failure.
- Do not label recovery unless a failure occurred first.
- Multiple failures may occur in a sequence, but only **one recovery** should be labeled per failure sequence.
- If a video contains only unsuccessful recovery attempts and no successful outcome, it must be rejected.

Failure and recovery should correspond to the same action. For example, if the operator fails while folding socks (Failure = Yes), then the recovery attempt should be the same action, meaning the next “folding socks” action should be marked as recovery if it’s successful.

Step-by-Step Instructions

Step 1 - ● **Team** Labels

The ● **Team** label is required and must always be set to the correct color for the system to accept your work. If you are an Alignerr or have not been assigned to a team select “other”.

Step 2 - ● **Bad Video** Labels

The ● **Bad Video** label is for videos that have issues with either the execution of the task, or the recording itself. You should start by watching the video from start to finish before you begin labeling ● **Actions.** As you become accustomed to the workflow, this will help you catch bad videos far sooner. Consider the following requirements:

1. The video must be in-focus and clear.
2. Hands performing a relevant ● **Action** and the object being manipulated must remain > 90% in frame at all times. The manipulation of objects must be clearly visible.
3. The video plays from start to finish without failed playback or other issues.
4. The operator doesn’t perform any ● **Actions** that are unrelated to the task.

5. The operator should complete the task in a natural manner. They shouldn't be moving very slowly, keep their arms stiff throughout the entire task, or move in a robotic manner.
6. During this step on EgoDiverse projects, you need to verify the ArUcO markers are in the correct position. See the FAQ at the end of these instructions for more information.

If you encounter a video that fails to meet these requirements:

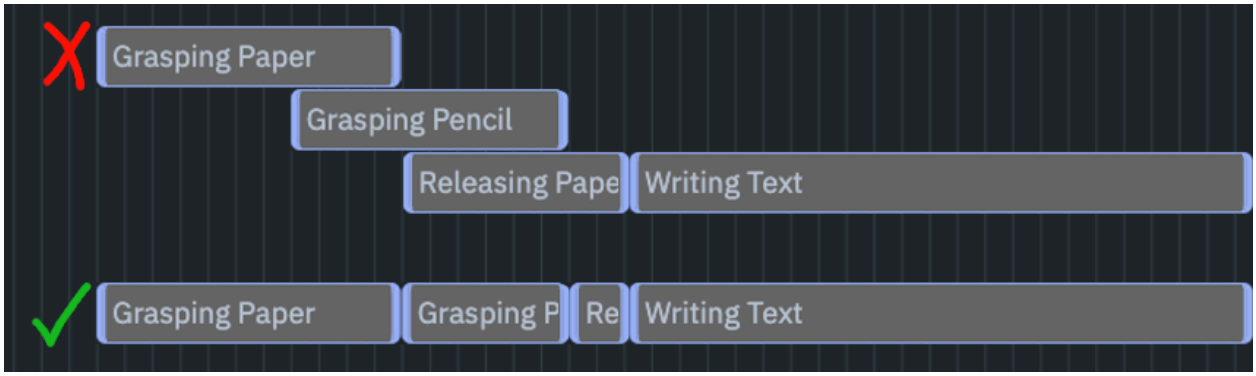
1. Tick the ● **Bad Video** checkbox
2. Add an ● **Action** label at the beginning of the task with a description of why the video is bad. Timestamps or frame numbers should be included whenever possible, for example "Frame 1160: Hand leaves frame while performing action." If you didn't catch the issue until after you started labeling, you can leave your existing labels and add the ● **Action** label to the frames that make the video bad.

Step 3 - ● **Action** Labels

Start the video and find the first ● **Action** the operator completes with their hands. For example, if the task is "cutting apple slices", the first ● **Action** you see may be the operator picking up an apple. The label should start when the operator starts to approach the apple with their hand(s). It should end when the operator touches the apple. The next ● **Action** label should begin on the next frame.

The description on ● **Action** labels should take the form of "<verb>ing <noun>" and be concise, consistent, and specific. For picking up and putting down objects, use the verbs "Grasping" and "Releasing" respectively. For the object, "Apple" is the correct level of granularity. You should not be overly specific (i.e. Granny Smith Apple) or overly Generic/vague (i.e. Fruit). You should avoid adjectives unless they are necessary. For example, in the "Cutting Apple Slices" task, the color of the apple is irrelevant and has no bearing on how the task is completed. However, in some tasks, such as "Sorting Bricks by Color", the description "Grasping Blue Brick" is directly relevant to the task.

Labels cannot overlap. If the hands perform two distinct ● **Action**s at the same time, the ●**Action** that started first should take priority over the latter:



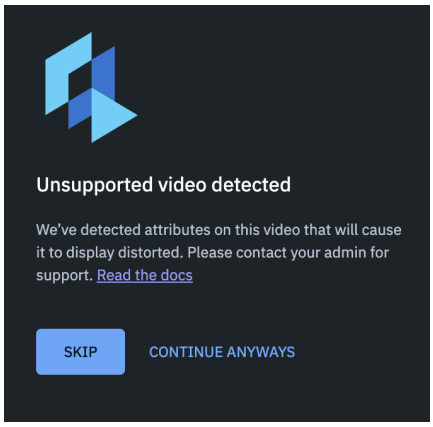
Frequently Asked Questions

When a task loads, I see “Unsupported Video Detected,” what should I do?

First, just try refreshing the page. This can be caused by connectivity issues.

If refreshing doesn’t work, click “Continue Anyways” to attempt to load the video. If it loads, play through the video once to make sure it isn’t corrupted.

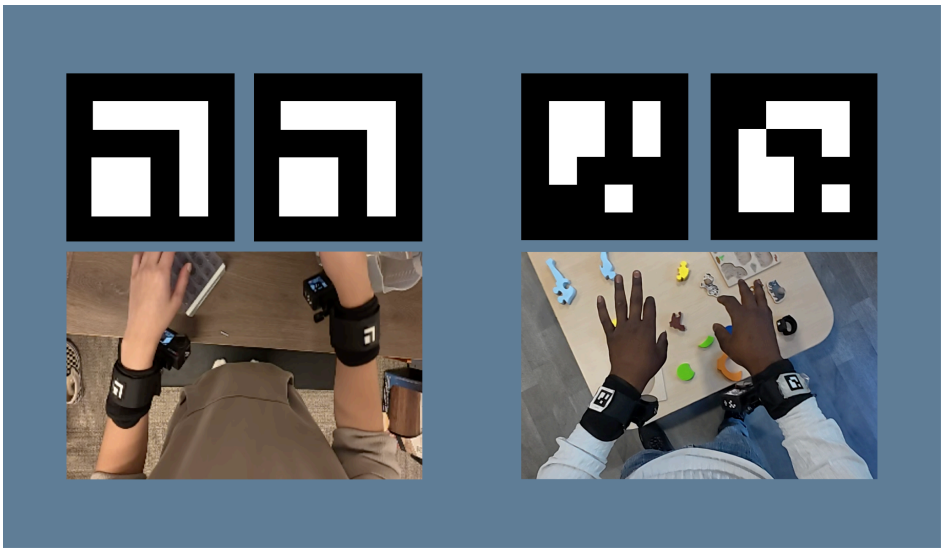
Iif plays through successfully and the video quality looks similar to others you have seen, label it as normal. If the video doesn’t load after clicking “Continue Anyways”, please skip the video.



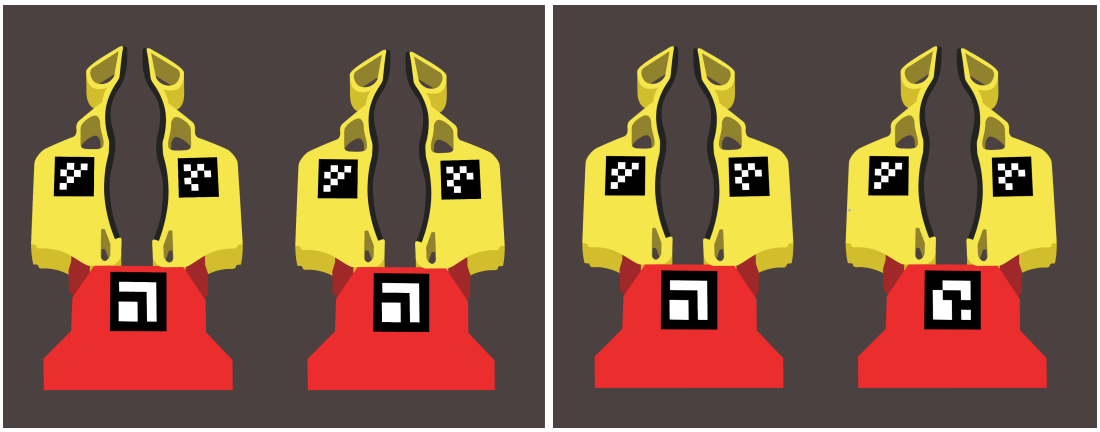
What do I need to check on the ArUcO markers for EgoDiverse projects?

On EgoDiverse projects, you should verify that the ArUcO markers are present and are oriented correctly. This includes rotation and left/right handedness.

For projects where the subject uses their ***hands*** to do the task, either of these two configurations are accepted:



For projects where the subject uses ***UMI grippers*** to complete the task, either of these two configurations are accepted:



Are There any Additional Looms or Other Video Examples?

1. Loom 1
2. Loom 2